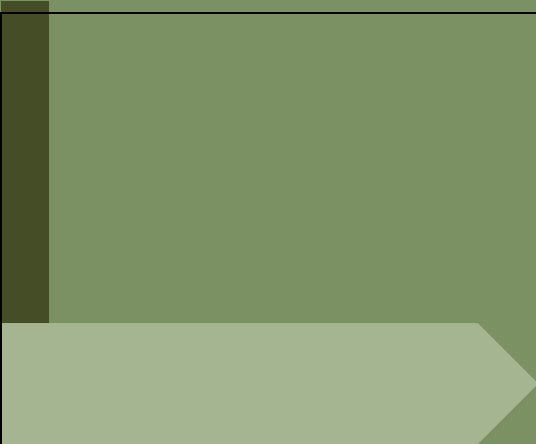
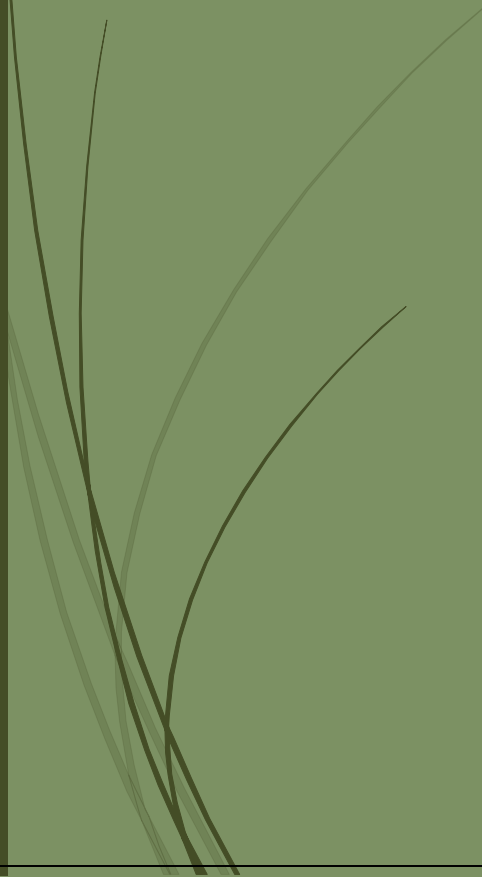




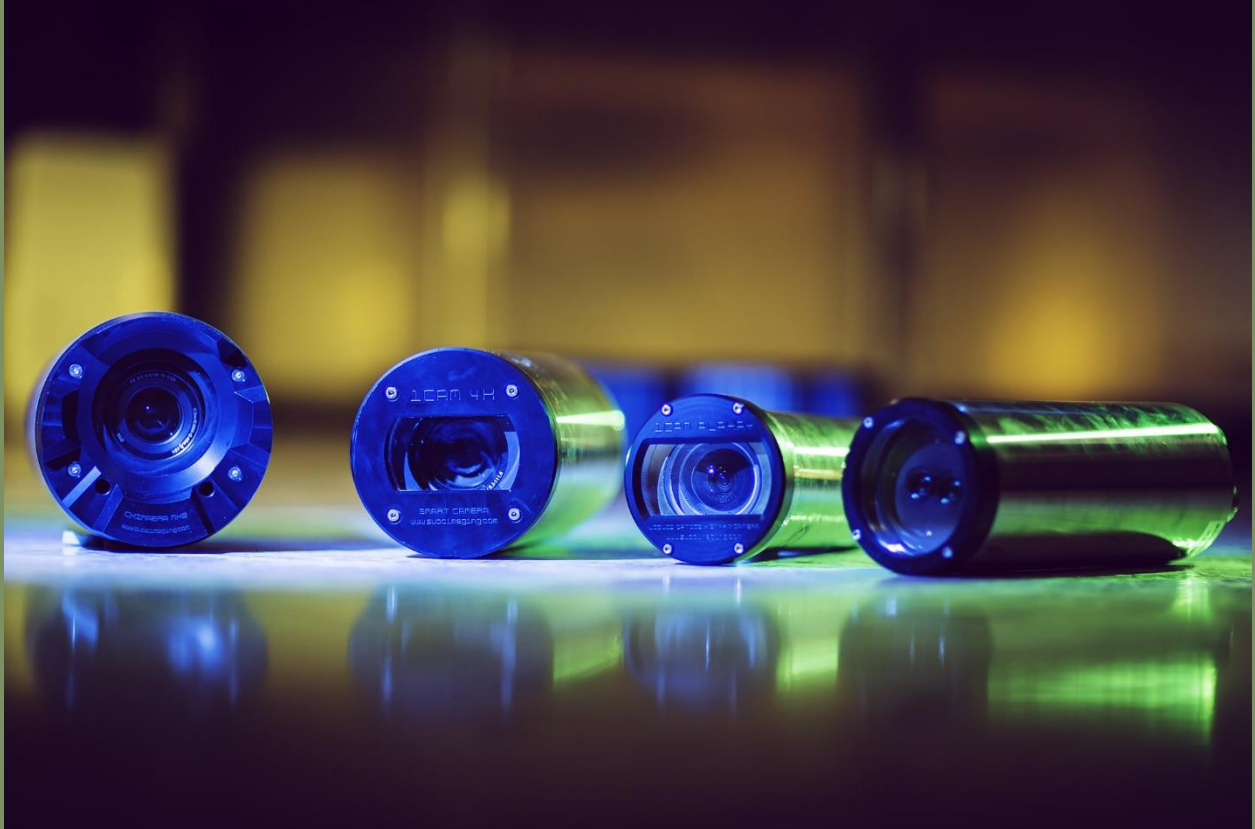
ROV '25 CERNON

[Cameras & Servos]



Presented by: Abdelrahman Ashraf

ROV's Camera & Servos :-



Camers : -

➤ 1 Cam Mk5

All models include built in **1080p HD Video** recording, **6-24 megapixel** stills and **608GB** of storage. Except the Lite, which is a low cost option



for live video only.

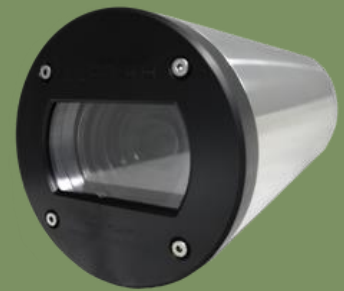
All models include built in **1080p HD Video** recording, **6-24 megapixel** stills and **608GB** of storage. Except the Lite, which is a low cost option for live video only.



➤ 1 Cam 4K

. It has a large 1" sensor to enable great low light footage, even in 4K resolution. If the system does not have a fiber optic link, video and stills can be recorded to **512GB** of storage.

Ethernet downloading and control, **SD video long line driver**, **100% Titanium housing**, auxiliary I/O pins for custom events and triggers and more

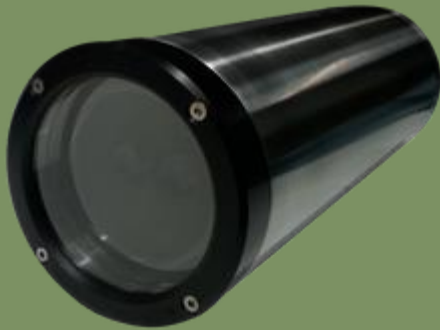


Designed as a 4K UHD **18x zoom** camera for systems equipped with fiber optics links, providing **live 2160/30p video** topside with the low latency inherent of fiber optics. Live SD and HD video is also available, as well as **20 megapixel digital stills** and internal **HD recording** as a backup



➤ 1 Cam 3D

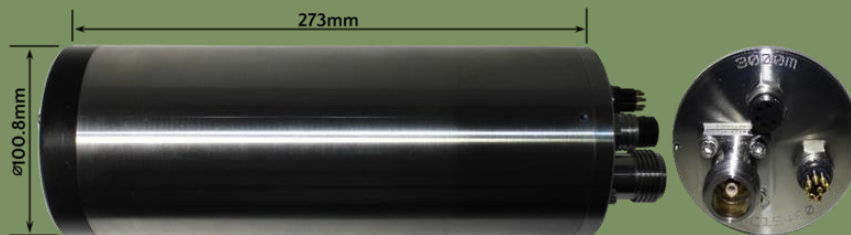
The 1Cam 3D utilizes two Full HD sensors to capture **true 3D 1080** video which is transmitted over Coax or Fiber optics.



In case of missing fiber infrastructure, SD, HD and 3D video is recorded to **512GB** storage allowing for well over **48 hours** of recording.

Designed as a main 3D HD color **10x zoom** camera where live video is transmitted over coax as 3G-SDI. Live SD video is also available.

The 1Cam 3D can take 6-20 Megapixel **digital stills** and has **internal recording** of 3D HD to 512GB of storage.



Availability and Financial Feasibility :

> "These types, however, are not commonly available in Egypt, and their cost exceeds our current budgetary capabilities."

Exploring Local Alternatives : -

IP cameras are digital surveillance cameras that work over a computer network (Ethernet).

- Connected via network cable (RJ45 / CAT5e/6) with power.
- Stream video directly in digital format.
- Can be viewed on a computer, monitor, or smartphone.
- **Advantages of IP Cameras:**
- High resolution (Full HD or 4K).
Live streaming without intermediaries.
Long distance operation (up to 100m and more with PoE extenders).
- Easy installation: power + data in one cable (PoE).

Compact, affordable compared to marine-grade cameras.

<> Disadvantages:

- > Require strong housing for underwater use.
- > Designed for rain/dust (IP67), not full immersion.
- > Need protection against high water pressure (in depth).

1. Hikvision DS-2CD1023G0-I

Type: Fixed IP Camera (Ethernet), 2 Megapixels.

Advantages:

- Full HD (1920×1080) good quality.
- Weatherproof and dustproof (IP67).
- Works with PoE or standard power supply.
- Live stream on the network (ONVIF compatible).
- Affordable and widely available in Egypt.

Disadvantages:

- Requires strong housing for underwater use (IP67 is not enough for immersion).
- Fixed lens – cannot be controlled remotely.
- Cannot withstand water pressure at depth.
- **Best for:** Budget live stream + shallow depth (with proper housing).

Approx. price: 1300-1700 EGP.

Places to buy:

- Hikvision official distributors in Cairo (Abdel Aziz Street)
- Noon Egypt (if available)



2. Dahua IPC-HDW1230T1-A-S5

Type: Fixed IP Camera (Ethernet), 2 Megapixels.

Advantages:

- Full HD with good image quality.
- IR night vision up to 30 meters.

- Weatherproof (IP67).
- Live stream + PoE support.
- Very good price.

Disadvantages:

- Same as Hikvision: needs housing for underwater use.
- Fixed lens.
- Cannot withstand pressure at depth.

Approx. price 1400–1800 EGP

Best for: Affordable live stream in simple projects.

Places to buy:

- Noon Egypt (online)
- Dahua official distributors in Cairo (Abdel Aziz Street, Cairo – CCTV/security stores)



3. Hikvision DS-2CD1043G0-I

Type: Fixed IP Camera (Ethernet), 4 Megapixels.

Advantages:

- Higher resolution (2560×1440) than the 2MP models.
- IR night vision.
- Weatherproof (IP67).
- Live stream + PoE support.

Disadvantages:

- Same as above: needs proper housing for underwater use.
- More expensive.
- Cannot handle underwater pressure.

Approx. price: 2200-2800 EGP.

Best for: Higher quality live stream in budget-friendly projects.

Places to buy:

- Hikvision official distributors in Cairo (Abdel Aziz Street)
- Noon Egypt (if available)

Finally :

I suggest number 2 : **2** Dahua

IPC-HDW1230T1-A-S5

Servos : -

What is Servo Motor?

The servo motor is a rotary actuator or linear actuator . It allows for precise control of angular or linear position, velocity and acceleration. It consists of a suitable motor coupled to a sensor for position feedback. It also requires a relatively sophisticated controller. The working principle of the servo motor and electromagnetic motor are same except the structure and the function are dissimilar. The standard servo motor uses the plastic gear whereas the high power servo motor uses the metal gear.



SG90



MG995



MG90

Plastic Gear Servos:

➤ SG90 :

- . SG90 (Tower Pro) – one of the most popular and cheapest.
- Torque: about 1.8-2.5 kg·cm.
- Very lightweight.
- Best for light cameras and small robots.
- 50-80 EGP per piece

Metal Gear Servos:

➤ MG90S

- .MG90S (Tower Pro) – same size as SG90 but with metal gears.

- Torque: about 2.2-2.5 kg·cm.
- Stronger and more durable than plastic.
- 90-150 EGP per piece

➤ **MG996R**

- Torque: around 10 kg·cm.
- Excellent for heavier cameras or robots in rough conditions.
- ~150–250 EGP per piece

Finally :

➤ I suggest number 3 : **MG996R**

➤ **Places to buy:**

Cairo - Abdel Aziz Street

- You'll find it at the bigger shops that sell robotics parts & drones.
- **Future Electronics Egypt :**

Address:

2 Sarayat Street, Abbassia, Cairo — **Across from Ain Shams University Faculty of Engineering**, above Abbassia Post Office.

Online

- Noon and Jumia — search for “MG996R”.

References : -

<https://robu.in/types-of-servo-motor/>

[Dahua Datasheet PDF](#)

[Hikvision Product Page](#)

[Hikvision Datasheet PDF](#)

<https://youtu.be/vbAZdQGTA0s?si=JnuOhh1mgIMYDYlW>

<https://www.rovinnovations.com/subc-imaging-underwater-cameras.html>